

Day: Probability Revision Sheet #2 Date: (1)

* Counting Sample Point

- 1) Random Experiments: Any process that can be repeated under identical conditions having unpredictable outcome.
- 2) Trial: A single iteration of a random experiment producing exactly one outcome.
- 3) Sample Space: The complete set of all possible outcomes of a random experiment.
- 4) Sample Points: A single outcome of a random experiment.
- 5) Outcome: A specific, single result of a random experiment.
- 6) Events: Any specific outcome or collection of outcomes of a random experiment. Representing a subset of sample space.

- Types of Events

- i) Simple Event: One Sample Point
- ii) Compound Event: More than one sample points
- iii) Null Event: No Sample Points (can never happen)

- Theorem 2.1

If one operation can be done in n_1 ways, and a second in n_2 ways, then both together can be done in $n_1 \times n_2$ ways.

Ex: 2 Coins

$$n_1 \times n_2 = n_1 n_2$$
$$2 \times 2 = 4 \text{ total outcomes}$$

H T H T

$$S = \{HH, HT, TH, TT\}$$

One Coin: One Die

$$n_1 \times n_2 = n_1 n_2$$

$$2 \times 6 = 12$$

H T 1, 2, 3, 4, 5, 6

$$S = \{H_1, H_2, \dots, H_6, T_1, T_2, \dots, T_6\}$$

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Q) In how many ways can 3 digit numbers be formed from:
0, 1, 2, 3, 4, 5, 6

a) Repetition allowed.

$$6 \times 7 \times 7$$

$$= 294$$

b) Repetition not allowed

$$6 \times 6 \times 5$$

$$= 180$$

c) Even numbers and repetition not allowed

$$\begin{array}{cccc} \text{E} & \text{E} & \text{O} & \text{E} \\ \hline 3 & & 1 & 3 \end{array}$$

$$6 \times 5 \times 1 + 5 \times 5 \times 3$$

$$30 + 75$$

$$= 105$$

d) Even Numbers, repetition allowed

$$6 \times 7 \times 4$$

$$= 168$$

e) Odd Numbers, Repetition not allowed

$$5 \times 5 \times 3$$

$$= 75$$

- Formulas

1) When Repetition is not allowed & no zero

$${}^n P_r = \frac{n!}{(n-r)!}$$

2) When Bache = Seats

$${}^n P_n = n!$$

3) When repetitions are allowed & no zero

$${}^n P_r = n^n$$

4) When Circular Arrangement

$${}^n P_r = (n-1)!$$

Alphabetic Repetition

$$\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1! \times n_2! \times \dots \times n_r!}$$

b) Combinations

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

* Permutations:-

- Use When Order Matters

• Arranging, ranking, forming numbers

* Combinations

• Use when order doesn't matter

• Selecting, choosing, picking a group.

* Probability

- Types of Events

1) Equally Likely Event

• Every event probability is same

$$P(A) = \frac{\text{Favourable Case}}{\text{Total Cases}}$$

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$A = \text{Even} = \{2, 4, 6\}$$

$$B = \text{Odd} = \{1, 3, 5\}$$

$$P(A) = \frac{3}{6} = 0.5$$

$$P(B) = \frac{3}{6} = 0.5$$

2) Exhaustive Event

Also called Sample Space

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3) Mutually Exclusive Event

- No common ^{element} between $\geq$ events

$$P(A \cup B) = P(A) + P(B)$$

4) Not Mutually Exclusive Event

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

5) Independent Event

$$P(A \cap B) = P(A) \cdot P(B)$$

6) Dependent Event

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

* Probability of A given that B happened

* Probability of letters ahead of j means before it. (a to i)

- Mutually Exclusive Events Formula

$$1. P(A) + P(A') = 1$$

$$2. P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$3. P(A \cap B') = P(A) - P(A \cap B)$$

$$4. P(A' \cap B) = P(B) - P(A \cap B)$$

$$5. P(A \cap B \cap C) = P(A \cap B) - P(A \cap B \cap C)$$

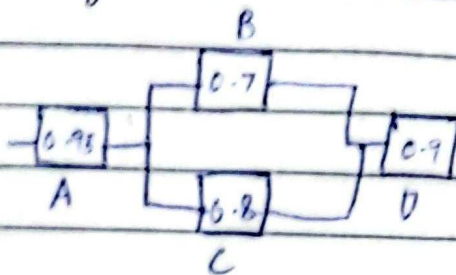
$$6. P(A \cap \bar{B} \cap C) = P(A) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C)$$

$$7. P(\bar{A} \cap B \cap C) = P(B) - P(A \cap B) - P(B \cap C) + P(A \cap B \cap C)$$

$$8. P(\bar{A} \cap \bar{B}) = 1 - P(A \cup B)$$

$$9. P(A \cap B) = P(A) \cdot P(B)$$

Circuit Diagrams



Probability in box is probability of working

$P(\text{system works})$ needs

$$A \rightarrow B \rightarrow D$$

$$A \rightarrow C \rightarrow D$$

$$A \rightarrow [B \cup C] \rightarrow D$$

$P(B \cup C)$ covers either and both

$$\text{so } P(\text{system works}) = P(A \cap P(B \cup C) \cap P(D))$$

$$= P(A) P(B' \cap C') P(D)$$

$$= P(A) [1 - (P(B')P(C'))] P(D)$$

* We convert union to intersect to use independent event properties

$$P(B' \cap C') = P(B') \cdot P(C')$$

* For Union between complex statements

$$P(A \cap B \cap C) \cup P(D \cap E \cap F)$$

$$\text{we set them as } P(X) \cup P(Y)$$

$$\text{and solve using } P(X) \cup P(Y) = P(X) + P(Y) - P(X \cap Y)$$

$$= P(X) + P(Y) - P(X) \cdot P(Y)$$

★ Bayes Theorem

Used when more than one condition for probability

$$P(M_1) = 0.2$$

$$P\left(\frac{D}{M_1}\right) = 0.01$$

$$P(M_2) = 0.3$$

$$P\left(\frac{D}{M_2}\right) = 0.02$$

$$P(M_3) = 0.5$$

$$P\left(\frac{D}{M_3}\right) = 0.03$$

$$P\left(\frac{M_1}{D}\right) = ? \quad P\left(\frac{M_1}{D}\right) = \frac{P(M_1 \cap D)}{P(D)}$$

$$\text{Since } P\left(\frac{D}{M_1}\right) = \frac{P(D \cap M_1)}{P(M_1)}$$

$$P(D \cap M_1) = P\left(\frac{D}{M_1}\right) P(M_1)$$

$$\text{and } P(D) = P(M_1 \cap D) + P(M_2 \cap D) + P(M_3 \cap D)$$

$$P(D) = P\left(\frac{D}{M_1}\right) P(M_1) + P\left(\frac{D}{M_2}\right) P(M_2) + P\left(\frac{D}{M_3}\right) P(M_3)$$

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$$P\left(\frac{M_1}{D}\right) = \frac{P(M_1 \cap D)}{P(D)} = \text{Ans}$$

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Chapter #3 Random Variable

- Random Variable is a function that associates a real number with each element in the sample space
- Discrete Random Variable
- Continuous Random Variable

* Discrete Random Variable

① $P(X = x_i) = f(x_i)$

$F \rightarrow$ combined Probability

- f^* shows individual probability
- also called probability density function (pdf) or probability distribution function or probability mass function.

② $0 \leq f(x_i) \leq 1$

③ $\sum_{i=1}^n f(x_i) = 1$

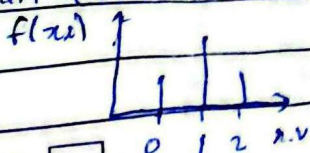
* Probability Distribution Table

X	$P(X = x_i) = f(x_i)$

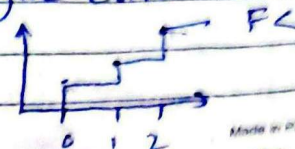
* Cumulative Frequency Distribution Table (In Probability we use $F <$ type)

X	$f(x_i) = P(X = x_i)$	$F(x) = P(X \leq x_i)$
0		

* Histogram (Drawn in Discrete Form)



* Ogive Curve



Expected Value

$$(1) E(X)^n = \sum_{i=1}^n X^n P(X = x_i)$$

$$\star E(X)^n = \sum_{i=1}^n x^n f(x_i)$$

$$(2) E(K) = K$$

$$(3) aE(x) = E(ax)$$

(4) If two random variables then

$$E(X \pm Y) = E(X) \pm E(Y)$$

(5) If two independent variable then

$$E(XY) = E(X)E(Y)$$

(6) $V(X) = E(X^2) - E(X)^2$ ← This formula is used in Random Variable

~~$\mu_2 = \mu_1^2$~~

$\mu_2 = \mu_1' - (\mu_1')^2$ ← This Formula is used in stats

$$(Q) E(4x+6) = 4E(4x) + E(6)$$

$$= 4E(x) + \cancel{E(6)} 6$$

$$(7) V(2x) = 2^2 (V(x))$$

and $V(4) = 0$

$$(Q) V(32x+3) = 9V(x) + 0$$

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$$E(X) = \sum x P(x)$$

$$\mu_2 = E(X^2) = E(X^2) - (E(X))^2$$

$$\mu_3 = E(X^3) - 3E(X^2)E(X) + 2E(X)^3$$

$$\mu_4 = E(X^4) - 4E(X^3)E(X) + 6E(X^2)E(X)^2 - 3E(X)^4$$

$$\sqrt{\beta_2} = \frac{\mu_3}{(\mu_2)^{3/2}}$$

$$\beta_2 = \frac{\mu_4}{(\mu_2)^2}$$

★ Binomial Distribution

$$b(x; n, p) = {}^n C_x p^x q^{n-x}$$

where $x = 0, 1, 2, \dots, n$

$$\text{Mean} = \mu = E(X) = np$$

$$\sigma^2 = \text{Variance} = V(X) = npq$$

$$\sigma = \text{SD} = \sqrt{npq}$$

$$\sum (p+q)^n = 1$$

- Bernoulli's Trial

You apply binomial distribution when Bernoulli's trial conditions are met.

3 conditions:

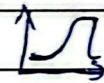
- Only 2 outcomes
- The probability of success is constant
- The trials are independent.

- Conditions of Binomial Experiment

- Fixed number of trials
- Only 2 possible outcomes (success or failure)
- Probability of success must be same for each trial.
- The trials are independent
- There are 2 parameters n & p in binomial distribution
- Variance is always less than mean. its mean
- When n is very large & p or q is very small, Binomial Distribution tend to Poisson Distribution.
- When n is very large & p or q is not very small, Binomial distribution tends to Normal Distribution.

* When p & q are equal $\therefore p = q = \frac{1}{2}$ the BD is symmetrical distrib

When $p > q$, it is negatively skewed



When $p < q$, it is positively skewed



* Binomial

Identification: You flip exactly n times, what's the probability you get exactly x success?

$$b(X; n, p) = \binom{n}{x} p^x q^{n-x}$$

$$b(X; n, p) = {}^n C_x p^x q^{n-x}$$

• Applicable for $n < 30$

$$E(x) = np$$

$$V(x) = npq$$

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★ Negative Binomial Distribution

Identification: You flip infinitely, what's the probability that your K^{th} success lands exactly on x^{th} flip

$$b^*(X; K, p) = \binom{x-1}{K-1} p^K q^{x-K}$$

$$b^*(X; K, p) = \binom{x-1}{K-1} p^K q^{x-K}$$

★ Geometric Distribution

Identification: You flip infinitely, what's the probability that your K^{th} success lands on exactly the x^{th} flip

$$g(X; p) = p q^{x-1}$$

★ Poisson Distribution

• Used where time period is involved. What's the probability exactly K events occur?

$$P(X; \lambda t) = \frac{e^{-\lambda t} (\lambda t)^x}{x!} \quad \text{where } \lambda t = \text{avg } X \text{ per period}$$

★ Hypergeometric & Multivariate Hyper Geometric Distribution

Solved using:

$$P(X) = \frac{P(\text{Outcomes})}{P(\text{Total})} \quad \text{and } P \text{ also calculated using } {}^nC_x$$

★ Binomial to Poisson Approximation

$n > 30$ and $p < 0.005$ then use approximation

$$np = \lambda \quad \text{and } b=1$$

$$\text{so: } \frac{e^{-\lambda} (\lambda)^x}{x!}$$

* Continuous Random Variable

* Integration gives Area under curve.

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Used to verify PDF (Cumulative Frequency $F(x)$)

* Also used to calculate unknown constants



* We can adjust limits to get area of a region

* Extra Question case is when data is given for X but you need values for X .

$$Q) f(x) = \begin{cases} x & 0 < x < 1 \\ 2-x & 1 \leq x < 2 \\ 0 & \text{e.w.} \end{cases}$$

$$Y = 60X^2 + 39X$$

Find mean of X

Solution:

$$E(Y) = E(60X^2 + 39X)$$

$$E(Y) = 60E(X^2) + 39E(X)$$

Formulas:

$$E(X) = \int_{-\infty}^{\infty} xf(x) dx$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$V(X) = E(X^2) - (E(X))^2$$

$$\sigma = \sqrt{V(X)}$$

$$F(x) = \int_{-\infty}^x f(x) dx$$

$$P(3 \leq x \leq 4) = F(4) - F(3)$$

* Normal Distribution

* Use table A3 to get values of $P(Z < 1.3)$
and:

$$P(Z > 1.3) = 1 - P(Z < 1.3)$$

If X is given instead of Z

$$P(X \leq 1.3) = P\left(Z < \frac{X - \mu}{\sigma}\right) = P\left(Z < \frac{1.3 - \mu}{\sigma}\right)$$

$$\text{then } Z = \frac{X - \mu}{\sigma}$$

then replace X with Z in $P(X < 1.3)$
and use table A3.

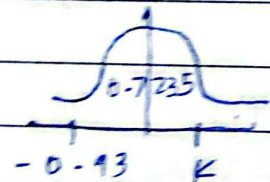
$$Q) P(-0.93 < Z < K) = 0.7235$$

$$P(Z < K) - P(Z < -0.93) = 0.7235$$

$$P(Z < K) = 0.8997$$

using table A3:

$$K = 1.28$$



$$Q) P(X < K) = 0.2236$$

$$P\left(X < \frac{K - \mu}{\sigma}\right) = 0.2236$$

$$Z = -0.76 \text{ using table A3 at } 0.2236$$

$$Z = \frac{X - \mu}{\sigma}$$

$$-0.76 = \frac{K - 18}{2.5}$$

$$K = 16.1$$

Q) $P(17 < X < 21)$

$$P\left(\frac{17 - \mu}{\sigma} < Z < \frac{21 - \mu}{\sigma}\right)$$

* Central Limit Theorem

- * Z is for normal distribution
- * X is for normal distribution when your data deviates from mean
- * Central Limit Theorem is when your sample size is small and not near normal distribution (and may have skewed)
- * When $n > 30$, the sampling distribution is approximately normal.

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \quad \text{where } \bar{X} = \text{sample mean}$$

If $P(Z < -0.597)$ we round to value in table which is closest i.e. 0.60.

$$\begin{aligned} \text{Q) } P(\bar{X} \geq 83) &= P\left(Z \geq \frac{83 - 75}{\frac{5}{\sqrt{5}}}\right) \\ &= 1 - P(Z < 3.57) \end{aligned}$$

* Normal Approximation to Binomial Distribution

$$\mu = np \quad \sigma = \sqrt{npq} \quad \sigma^2 = npq$$

$$Z = \frac{X - np}{\sqrt{npq}}$$

$$S = \sqrt{\frac{(\bar{x} - \bar{x})^2}{n-1}}$$

$$\sigma = \sqrt{\frac{(x - \bar{x})^2}{n}}$$

* Binomial is discrete but Normal is continuous so we need to correct the inequalities

Binomial	With Correction
$P(X=150)$	$P(149.5 < X < 150.5)$
$P(X \leq 150)$	$P(X < 150.5)$
$P(X < 150)$	$P(X < 149.5)$
$P(X \geq 150)$	$P(X > 149.5)$
$P(X > 150)$	$P(X > 150.5)$

$$\begin{aligned}
 Q) P(120 < X < 150) &= P(120.5 < X < 149.5) \\
 &= P\left(\frac{120.5 - np}{\sqrt{npq}} < Z < \frac{149.5 - np}{\sqrt{npq}}\right) \\
 &= P
 \end{aligned}$$

* Syntax for Qs

$$P(25 \leq X \leq 30) = \sum_{x=25}^{30} b(x; 80, 0.25) \approx P(1.11 < Z < 2.7)$$

* Exponential Distribution

$$f(x; \beta) = \begin{cases} \frac{1}{\beta} e^{-\frac{x}{\beta}} & x > 0 \\ 0 & \text{e.w.} \end{cases}$$

where $\mu = \beta$ $\sigma^2 = \beta^2$

Q) $P(X > 5)$ when $\beta = 3$

$$\begin{aligned}
 P(X > 5) &= \int_5^{\infty} \frac{1}{3} e^{-x/3} dx \\
 &= -1 \left| e^{-x/3} \right|_5^{\infty} \\
 &= -1 (0 - e^{-5/3}) \\
 &= e^{-5/3}
 \end{aligned}$$

$$P(X < 3) = \int_0^3 \frac{1}{4} e^{-x/4} dx$$

* Statistical Inference → getting data about population from sample

↳ 2 Types

① Statistical Estimation

↳ ① Point Estimation → a single value estimate of a population

↳ ② Interval Estimation → range of values within which the values of population is expected to lie

↳ Z-test

↳ t-test

* Confidence Interval

* Case # 1) σ Known & $n \geq 30$, $n < 30$

$$\bar{X} - Z_{\frac{\alpha}{2}} \left(\frac{\sigma}{\sqrt{n}} \right) < \mu < \bar{X} + Z_{\frac{\alpha}{2}} \left(\frac{\sigma}{\sqrt{n}} \right)$$

* Case # 2) σ unknown & $n \geq 30$

$$\bar{X} - Z_{\frac{\alpha}{2}} \left(\frac{S}{\sqrt{n}} \right) < \mu < \bar{X} + Z_{\frac{\alpha}{2}} \left(\frac{S}{\sqrt{n}} \right)$$

* Case # 3) $n < 30$, σ unknown

$$\bar{X} - \left(t_{\frac{\alpha}{2}, n-1} \right) \left(\frac{S}{\sqrt{n}} \right) < \mu < \bar{X} + \left(t_{\frac{\alpha}{2}, n-1} \right) \left(\frac{S}{\sqrt{n}} \right)$$

For $\left(\frac{t_{\alpha}}{2}, n-1\right)$ use table A4

For $Z_{\frac{\alpha}{2}}$ use table A3

Population		Sample	
Mean	μ	Mean	$\bar{x}$
Size	N	Size	n
Var	σ^2	Var	S^2
Population SD	$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n \Rightarrow N}}$	Sample SD	$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$

* In confidence interval, -ve sign of $Z_{\frac{\alpha}{2}}$ value from table is dropped

* Hypothesis Testing

$H_0 \rightarrow$ Null Hypothesis

$H_1 \rightarrow$ Alternative Hypothesis

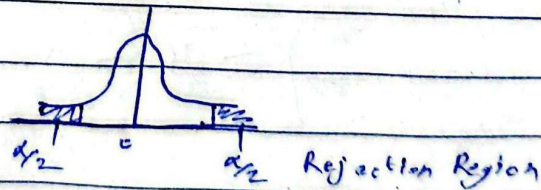
Level of Significance $\alpha \rightarrow$ If not given $\alpha = 0.05 \rightarrow$ Shows Rejection Region

① $H_0: \mu = \text{some value}$

vs $H_1: \mu \neq \text{some value}$

or $H_1: \mu < \text{some value}$ or $\mu > \text{some value}$

Two Tail Test



* Reject H_0 if

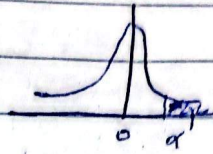
$$t_{cal} > t_{\frac{\alpha}{2}} \quad \text{or} \quad t_{cal} < -t_{\frac{\alpha}{2}}$$

* Reject H_0 if

$$Z_{cal} > Z_{\frac{\alpha}{2}} \quad \text{or} \quad Z_{cal} < -Z_{\frac{\alpha}{2}}$$

② $H_0: \mu \leq \text{some value}$ v/s $H_1: \mu > \text{some value}$

Upper Tail Test

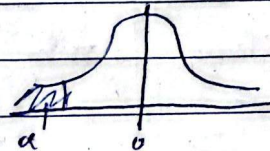


Reject H_0 if
 $Z_{cal} > Z_{\alpha}$

or $t_{cal} > t_{\alpha}$

③ $H_0: \mu \geq \text{some value}$ v/s $H_1: \mu < \text{some value}$

Lower Tail Test



Reject if
 $Z_{cal} < -Z_{\alpha}$

or $t_{cal} < -t_{\alpha}$

* Steps

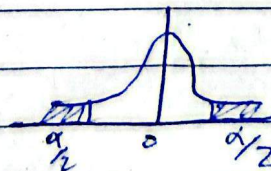
* Step #1: State Hypothesis

$$H_0: \mu = 5.5$$

$$H_1: \mu \neq 5.5$$

* Step #2: Level of Significance

$$\alpha = 0.05$$



* Step #3: Test Statistic

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

or $t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$

Step #4) Critical Region (State when Reject)

Reject H_0 if $Z_{cal} < -Z_{tab}$

or
Reject H_0 if $t_{cal} < t_{tab}$
where $t(\alpha, n-1) = t_{tab}$

Step #5) Calculation

Step #6) Decision

H_0	H_1	Mean Testing	Variance Testing
$=$	$\neq$	$< = -\alpha/2$	$< = 1 - \alpha/2$
	$<, >$	$> = \alpha/2$	$> = \alpha/2$
$\leq$	$>$	$> = \alpha$	α
$\geq$	$<$	$< = -\alpha$	$1 - \alpha$

* Chi-Square Distribution

2. Test for Homogeneity (Does data exceed variance?)

Step #1) State Hypothesis

$$H_0: \sigma^2 = 0.03$$

$$H_1: \sigma^2 \neq 0.03$$

$$\sigma^2 > 0.03, \sigma^2 < 0.03$$

Step #2) Level of Significance

$$\alpha = 0.01$$

Step #3) Test Statistic

$$\chi^2 = (n-1) \frac{S^2}{\sigma^2}$$

Step #4) Critical Region

Reject H_0 if $\chi^2_{cal} > \chi^2_{tab} = \chi^2_{\alpha/2, n-1}$

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or

Reject H_0 if $\chi^2_{cal} < \chi^2_{tab} = \chi^2_{1-\alpha/2, n-1}$

Step #5) Calculation

Step #6) Decision

Test for Independence

- χ^2 distribution is continuous & unimodal
- χ^2 distribution is always positive
- χ^2 distribution mean = v
- χ^2 distribution variance = $2v$
- For small v ($v < 10$) distribution is skewed right
- As degree of freedom increases, chi square distribution approaches ND & symmetry of curve increases

Step #1) State Hypothesis

H_0 : ~~Assumption~~ A & B are independent of each other

H_1 : A & B are dependent of each other.

Step #2) Level of Significance

$$\alpha = 0.05$$

Step #3) Test Statistics

$$\chi^2 = \frac{(O_i - E_i)^2}{E_i}$$

$$E_i = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$$

Step #4) Critical Region

Reject H_0 if $\chi^2_{cal} > \chi^2_{tab} = \chi^2_{\alpha, (r-1)(c-1)}$

Step #5) Calculations

Draw Tables

Observed	Expected	$(O_i - E_i)$	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
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* Regression

- Equation Types:

$$y = a + bx$$

$$y = a + bx + cx^2$$

$$y = ab^x$$

$$y = ax^b$$

$$y = ae^{xb}$$

} Log both sides before to make linear
coefficient multiplication & BS

} ln both sides to make linear

Q) For the given data check which equation is best fitted with the method of least square & calculate residual

$$y = a + bx \quad \text{--- (A)}$$

Multiply equation with coefficients of a & b then summation both sides

$$\sum y = \sum a + \sum bx$$

$$\sum y_i = na + x \sum x_i$$

$$\sum xy = \sum ax + \sum bx^2$$

$$\sum x_i y_i = a \sum x_i + b \sum x_i^2$$

Solve equations simultaneously for a & b to get equation

Use equation to get $\hat{y}_i$ (predicted y value)

$$\text{if } a = -128.14 \quad b = 62.7$$

$$\hat{y}_i = -128.13 + 62.7x_i$$

$\sum e_i$ gives error where $e_i = (y_i - \hat{y}_i)$

Solve using other equations and choose the one with lowest value of absolute $\sum e_i$